Ortho AI LLC
Jonathan Vigdorchik
Orthopaedic Surgeon
155 East 76th Street
Apt 2H
New York, NY 10021

January 2, 2025

Re: K241696

Trade/Device Name: Ortho AI
Regulation Number: 21 CFR 892.2050
Regulation Name: Medical Image Management And Processing System
Regulatory Class: Class II
Product Code: QIH
Dated: November 26, 2024
Received: November 27, 2024

Dear Jonathan Vigdorchik:

We have reviewed your section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (the Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database available at https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Additional information about changes that may require a new premarket notification are provided in the FDA guidance documents entitled "Deciding When to Submit a 510(k) for a Change to an Existing Device" (https://www.fda.gov/media/99812/download) and "Deciding When to Submit a 510(k) for a Software Change to an Existing Device" (https://www.fda.gov/media/99785/download).

Your device is also subject to, among other requirements, the Quality System (QS) regulation (21 CFR Part 820), which includes, but is not limited to, 21 CFR 820.30, Design controls; 21 CFR 820.90, Nonconforming product; and 21 CFR 820.100, Corrective and preventive action. Please note that regardless of whether a change requires premarket review, the QS regulation requires device manufacturers to review and approve changes to device design and production (21 CFR 820.30 and 21 CFR 820.70) and document changes and approvals in the device master record (21 CFR 820.181).

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR Part 803) for devices or postmarketing safety reporting (21 CFR Part 4, Subpart B) for combination products (see https://www.fda.gov/combination-products/guidance-regulatory-information/postmarketing-safety-reporting-combination-products); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR Part 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR Parts 1000-1050.

All medical devices, including Class I and unclassified devices and combination product device constituent parts are required to be in compliance with the final Unique Device Identification System rule ("UDI Rule"). The UDI Rule requires, among other things, that a device bear a unique device identifier (UDI) on its label and package (21 CFR 801.20(a)) unless an exception or alternative applies (21 CFR 801.20(b)) and that the dates on the device label be formatted in accordance with 21 CFR 801.18. The UDI Rule (21 CFR 830.300(a) and 830.320(b)) also requires that certain information be submitted to the Global Unique Device Identification Database (GUDID) (21 CFR Part 830 Subpart E). For additional information on these requirements, please see the UDI System webpage at https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/unique-device-identification-system-udi-system.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to https://www.fda.gov/medical-devices/medical-device-safety/medical-device-reporting-mdr-how-report-medical-device-problems.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance) and CDRH Learn (https://www.fda.gov/training-and-continuing-education/cdrh-learn). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (https://www.fda.gov/medical-devices/device-advice-comprehensive-regulatory-assistance/contact-us-division-industry-and-consumer-education-dice) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

Jessica Lamb

Jessica Lamb, Ph.D
Assistant Director
Imaging Software Team
DHT8B: Division of Radiological Imaging
Devices and Electronic Products
OHT8: Office of Radiological Health
Office of Product Evaluation and Quality
Center for Devices and Radiological Health

Enclosure

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

Indications for Use

Form Approved: OMB No. 0910-0120
Expiration Date: 07/31/2026
See PRA Statement below.

Submission Number *(if known)*

K241696

Device Name

Ortho AI

Indications for Use *(Describe)*

Ortho AI is an image-processing software indicated to assist in making measurements for a total hip arthroplasty, total knee arthroplasty, and lumbar spine fusion surgery.
It is intended to assist in the measurement of x-ray images by measuring lengths, angles and position of implants relative to the bone structures of interest provided, that the points of interest can be identified from radiology images.
The device allows for overlaying of digital annotations on radiological images and includes tools for performing measurements using the images and digital annotations. The software is not for primary image interpretation. The software is not for use on mobile phones.

Intended patient population: Adult patients >=22 years old, with appropriate imaging, undergoing primary hip replacement, primary knee replacement, and lumbar spine surgery.
Intended user population: orthopaedic surgeons who perform hip and knee replacement, and orthopaedic/neurosurgeons who perform lumbar spine surgery.

Type of Use *(Select one or both, as applicable)*

☒ Prescription Use (Part 21 CFR 801 Subpart D) ☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

K241696

510(k) Summary

Device Trade Name: Ortho AI

Manufacturer: Ortho AI
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Phone: 314-662-3222

Contact: Jonathan Vigdorchik
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Prepared By: MCRA, LLC
803 7th St NW
Washington, DC 20001
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Date Prepared: November 25, 2024

Device Trade Name: Ortho AI
- **Classification:** 892.2050 Medical image management and processing system
- **Class:** Class II
- **Product Code:** QIH

Predicate Device: **KOALA -** K192109
- **Classification:** 892.2050 Medical image management and processing system
- **Class:** Class II
- **Product Code:** QIH

Indications for Use:

Ortho AI is an image-processing software indicated to assist in making measurements for a total hip arthroplasty, total knee arthroplasty, and lumbar spine fusion surgery.

It is intended to assist in the measurement of x-ray images by measuring lengths, angles and position of implants relative to the bone structures of interest, provided that the points of interest can be identified from radiology images.

The device allows for overlaying of digital annotations on radiological images and includes tools for performing measurements using the images and digital annotations. The software is not for primary image interpretation. The software is not for use on mobile phones.

Intended patient population: Adult patients >= 22 years old, with appropriate imaging, undergoing primary hip replacement, primary knee replacement, and lumbar spine surgery.

Intended user population: orthopaedic surgeons who perform hip and knee replacement, and orthopaedic/neurosurgeons who perform lumbar spine surgery.

Device Description:
Ortho AI is a software as a medical device (SaMD) system that provides preoperative planning data for hip replacement surgery, knee replacement surgery, and lumbar spinal fusion surgery using AI/ML models that are semi-automated and interpretable. The software guides the user through a predetermined workflow that begins with the use of preoperative radiographic images as input to the software. As part of this initial preoperative workflow, the software places digital annotations on these preoperative images, which can be modified by the user (semi-automated). The software additionally includes functionality to store user, patient, and case information.

Software/firmware input
As a non-invasive software as a medical device (SaMD) system, Ortho AI requires, as input, preoperative radiographic images of the pelvis, knee, or spine. Input images may be one of a large number of standard image formats that are supported (.jpg, .bmp, etc.), a DICOM image. The image can also be uploaded intraoperatively from a fluoroscopic c-arm via a direct cable connection. Images must be uploaded by a healthcare professional.

Software/firmware output
The software output will be an HTML page that displays the processed image in JPEG format. The output displays the measurements of angles and length of anatomical structures that are essential for pre-operative planning for the anatomical site displayed in the image. The output can only be viewed by a healthcare professional.

Software functions
Ortho AI is a software-only device intended to be used by healthcare professionals for the preoperative and intraoperative planning of total hip replacement surgery, partial/total knee replacement surgery, or lumbar spine surgery. The software semi-automates templating/planning for surgery. In the standard of care pre-operative planning procedure, a healthcare provider templates manually or digitally on a radiograph by drawing lines, making angular measurements, and sizing implants using acetate or digital templates. The Ortho AI software will semi-automate the above listed tasks by taking a radiological image as input and delivering key measurements of the surgeon's interest through an AI algorithm. These landmarks are then editable by the user.

Substantial Equivalence:

Feature	**Subject Device**	**Predicate Device**	**Comparison**
510(k) Number	K241696	K192109	N/A
Name	Ortho AI	IB Lab's KOALA Software	N/A
Classification Name and Product Code	System, Image Processing, Radiological (QIH)	System, Image Processing, Radiological (LLZ)	Same

Runs on Server	Yes	Yes	Same
Image input	JPEG, PNG, BMP, DICOM	DICOM compliant images collected in other devices in either digitally computed (CR) or directly digital (DX) formats.	Similar. The subject device is validated to additionally analyze other standard image formats and does not raise new questions about safety or effectiveness.
Imaging processing	Landmark detection, user can edit landmark	Knee detection; Landmark detection; Joint space detection	Same
Anatomical Area	Knee, Hip, Lumbar Spine	Leg	Similar. The subject device is intended to be used on the knee, hip, and lumbar spine while the predicate device is only for the leg. This difference in specific anatomic locations does not raise new types of questions for safety or effectiveness.
Measurement	Length and angle	Length and angle	Same
Way of Measurement	Semi-automatic (AI driven)	Automatic (AI driven)	Similar. The subject device requires the user to review and edit the device output as necessary while the predicate device does not as a fully automatic device.
Intended User	Trained Professional	Trained Professional	Same
Indications for Use	Ortho AI is an image-processing software indicated to assist in making measurements for a total hip arthroplasty, total knee arthroplasty, and lumbar spine fusion surgery. It is intended to assist in the measurement of x-ray images by measuring lengths, angles and position of implants relative to the bone structures of interest, provided that the points of interest can be	IB Lab KOALA is a radiological fully-automated image processing software device of either computed (CR) or directly digital (DX) images intended to aid medical professionals in the measurement of minimum joint space width; the assessment of the presence or absence of sclerosis, joint space narrowing, and osteophytes based OARSI criteria for these parameters; and the presence or absence of	The subject device performs measurements of lengths and angles on hip, knee, and lumbar spine images. The predicate device performs length and angle measurements on leg images. This difference in specific anatomic locations does not raise new questions for safety or effectiveness and therefore does not induce changes in the intended use

	identified from radiology images. The device allows for overlaying of digital annotations on radiological images and includes tools for performing measurements using the images and digital annotations. The software is not for primary image interpretation. The software is not for use on mobile phones. Intended patient population: Adult patients >=22 years of age, with appropriate imaging, undergoing primary hip replacement, primary knee replacement, and lumbar spine surgery. Intended user population: orthopaedic surgeons who perform hip and knee replacement, and orthopaedic/neurosurgeons who perform lumbar spine surgery.	radiographic knee OA based on Kellgren & Lawrence Grading of standing, fixed-flexion radiographs of the knee. It should not be used in-lieu of full patient evaluation or solely relied upon to make or confirm a diagnosis. The system is to be used by trained professionals including, but not limited to, radiologists, orthopedics, physicians, and medical technicians.	

Performance Testing Summary:
Software Validation and Verification was performed.

<u>*Acceptance Criteria and Sample size*</u>
The overall aim was to achieve a Dice coefficient greater than 0.85 (*Dice* > 0.85).
A test sample size of ≥ 150 samples and observing an effect size of ≥ 0.01370, we can be confident achieving a statistical power of ≥ 80%.

Acceptance criteria for a mean Dice coefficient at a minimum of 0.85 for all algorithms and all connected domains.

<u>Model summary:</u>

Data independence: We ensured full data independence by adopting a patient-level partitioning process, where each patient contributed only a single x-ray image to the dataset, eliminating any risk of inter-sample dependencies or data leakage between the training, validation, and testing phases.

Truthing process: 3 fellowship-trained ABOS board-certified orthopaedic surgeons with each greater than 10 years of experience, followed a 2+1 truthing process where two blinded orthopaedic surgeons reviewed each segmented image in the testing set and applied modifications. After the reviews from each blinded surgeon, a final senior-level surgeon adjudicator reviewed the modifications and added further modifications to the segmentations, if necessary.

Hip model:
- 1,367 images from 1,367 patients
- Average age: 62
- Female: 52%
- Ethnicity: 78% white, 18% Black/African American, 1% Other (non-white), 3% Asian

Hip-spine model:
- 4,836 images from 4,836 patients
- Average age: 62
- Female: 61%
- Ethnicity: 78% white, 18% Black/African American, 1% Other (non-white), 3% Asian

Knee model:
- 4,536 images from 4,536 patients
- Average age: 61
- Female: 52%
- Ethnicity: 78% white, 18% Black/African American, 1% Other (non-white), 3% Asian

Standalone algorithm testing

Hip model:
- LLD measurements within +/- 1.96mm of human measurement
- Offset (global) within +/- 0.88mm of human measurement
- SFP angle within +/- 1.05mm of human measurement
- Overall performance as measured by Dice coefficient shows all connected domains to be above our acceptance criteria of 0.85.
- Subgroup analysis by x-ray machine also shows that all connected domains, across all x-ray machine types, to be above our acceptance criteria of 0.85.

Hip-spine model:
- SS, SPT, APPt, PI, LL, PI-LL – all within 2 degrees of human measurement
- No statistical difference between human vs. machine learning measurements
- Overall performance as measured by Dice coefficient shows all connected domains to be above our acceptance criteria of 0.85.
- Subgroup analysis by x-ray machine also shows that all connected domains, across all x-ray machine types, to be above our acceptance criteria of 0.85.

Knee model:
- LDFA , mPTA, aHKA, aJLOA all within 2 degrees of human measurement
- No statistical difference between human vs. machine learning measurements

- Overall performance as measured by Dice coefficient shows all connected domains to be above our acceptance criteria of 0.85.
- Subgroup analysis by x-ray machine also shows that all connected domains, across all x-ray machine types, to be above our acceptance criteria of 0.85.

Conclusion

Ortho AI demonstrates substantial equivalence to the predicate device. The subject device has the same intended use and principles of operation; furthermore, it has similar indications and technological characteristics as its predicate device. The minor differences between subject and predicate device in indications do not alter the intended use of the subject device and do not raise new or different questions regarding its safety and effectiveness when used as labeled. Performance data demonstrate that the device performs as intended. Verification and validation testing, including the standalone software performance test, supports the safety of the device and demonstrates that Ortho AI performs as intended. Therefore, Ortho AI demonstrates substantial equivalence to the predicate device.